

# Machine Learning and Data Mining - Lab 1

Probability, Expectation, Classification / Regression

$k$ -Nearest Neighbour

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## 1. Exercises - In Class

### 1.1 Probability and Expectation

**Exercise 1.1** Two discrete random variables  $X$  and  $Y$  taking the values  $\{0, 1, 2\}$  have a joint p.m.f. given by

|         | $X = 0$ | $X = 1$ | $X = 2$ |
|---------|---------|---------|---------|
| $Y = 2$ | 0       | 1/6     | 1/8     |
| $Y = 1$ | 1/8     | 1/6     | 1/8     |
| $Y = 0$ | 1/6     | 1/8     | 0       |

- (a) Compute  $P(X = 1, Y = 2)$  and  $P(X = 1 | Y = 2)$ .
- (b) Compute the conditional expectation  $E_P[X | Y = 2]$ .

**Exercise 1.2** Let  $X$  denote the size of a job. Suppose that

$$X = \begin{cases} 1 & \text{w/ prob. 0.1} \\ 2 & \text{w/ prob. 0.2} \\ 3 & \text{w/ prob. 0.3} \\ 4 & \text{w/ prob. 0.2} \\ 5 & \text{w/ prob. 0.2} \end{cases}$$

Let  $A$  be the event that the job is *small* meaning that its size is  $\leq 3$ . Our goal is to understand the conditional p.m.f. of  $X$  given event  $A$ .

- (a) Compute  $P(X = 1 | A)$ .
- (b) Compute  $P(X = x | A)$  if  $x \notin A$ .
- (c) Compute  $E_P[X]$  and  $E_P[X | A]$ .

**Exercise 1.3** A university wants to build a simple classifier for predicting whether an applicant will be admitted to a master program ( $C = 1$ ) or not ( $C = 0$ ), based on whether their GPA is *High* ( $X = 1$ ) or *Low* ( $X = 0$ ). Here is the historical data:

| Number of Observations | $X$ | $C$ |
|------------------------|-----|-----|
| 28                     | 0   | 0   |
| 3                      | 0   | 1   |
| 7                      | 1   | 0   |
| 12                     | 1   | 1   |

Treat relative frequencies as the true probabilities and derive the probabilities from the given table.

- (a) Compute  $P(C = 1)$ ,  $P(X = 1)$ ,  $P(C = 1 | X = 1)$  and  $P(C = 1 | X = 0)$ .
- (b) A natural decision rule is to predict the class  $c$  that maximizes  $P(C = c | X = x)$ . What would this classifier predict for an applicant with a High GPA? A Low GPA?

## 1.2 Classification vs Regression

**Exercise 2.1** For each description: classification, regression, or neither? Identify a reasonable input variable(s) and output variable(s).

1. We collect a set of data on the top 500 firms in the US. For each firm we record profit, number of employees, industry and the CEO salary. We are interested in understanding which factors affect CEO salary.
2. We are considering launching a new product and wish to know whether it will be a success or a failure. We collect data on 20 similar products that were previously launched. For each product we have recorded whether it was a success or failure, price charged for the product, marketing budget, competition price, and ten other variables.
3. We are interested in predicting the % change in the USD/Euro exchange rate in relation to the weekly changes in the world stock markets. Hence we collect weekly data for all of 2012. For each week we record the % change in the USD/Euro, the % change in the US market, the % change in the British market, and the % change in the German market.

## 1.3 $k$ -Nearest Neighbour ( $k$ -NN)

**Exercise 3.1** Let consider the following dataset, which contains brightness and saturation values, along with the corresponding color class:

| Brightness | Saturation | Class |
|------------|------------|-------|
| 40         | 20         | Red   |
| 50         | 50         | Blue  |
| 60         | 90         | Blue  |
| 10         | 25         | Red   |
| 70         | 70         | Blue  |
| 60         | 10         | Red   |
| 25         | 80         | Blue  |

- (a) Using the given dataset, apply the 1-NN and calculate the training accuracy. Explain why the training accuracy is always 100% when  $k = 1$ .
- (b) Consider a test data  $x$  with Brightness 20 and Saturation 35. Compute  $P(\text{Red} | x)$  and  $P(\text{Blue} | x)$  using 5-NN and Euclidean distance. Based on these probabilities, determine the predicted class.
- (c) Suppose that the utility associated with each possible decision and true class is given by the following utility matrix  $U(y, a)$ :

|                   | $a = \text{Red}$ | $a = \text{Blue}$ |
|-------------------|------------------|-------------------|
| $y = \text{Red}$  | 0                | -1                |
| $y = \text{Blue}$ | -2               | 0                 |

where  $y$  denotes the true class and  $a$  denotes the predicted class. Calculate the expected utilities  $E_P[U(a = \text{Red} \mid x)]$  and  $E_P[U(a = \text{Blue} \mid x)]$ .

- (d) Which predicted class  $a^*$  do maximize the expected utility? Explain why the class with the highest probability is not necessarily the class that maximizes expected utility.

## 2. Exercises - Extra

### 2.1 Probability and Expectation

**Exercise 1.1** Let  $X : \Omega \rightarrow \mathbb{R}$  be a discrete random variable,  $A \subset \mathbb{R}$  an event and  $P$  a probability measure. Show that  $E_P[E_P[X \mid A]] = E_P[X]$ .

**Exercise 1.2** Suppose that we independently roll two standard 6-sided dice. Let  $X_1$  be the number that shows on the first die,  $X_2$  the number on the second die and  $X$  the sum of the numbers on the two dice.

- (a) What is  $E_P[X \mid X_1 \text{ is even}]$ ?
- (b) What is  $E_P[X \mid X_1 = X_2]$ ?
- (c) What is  $E_P[X_1 \mid X = 9]$ ?

**Exercise 1.3** Show that  $\mathbb{E}[(X - c)^2] = \text{Var}(X) + (\mathbb{E}[X] - c)^2$  for any constant  $c$  and random variable  $X$ . (Conclude that the best constant predictor is  $c^* = \mathbb{E}[X]$  with error  $\text{Var}(X)$ )

**Exercise 1.4** Show that if  $X$  and  $Y$  are independent random variables, then

- (a)  $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y)$ .
- (b)  $E_P[X \mid Y] = E_P[X]$

### 2.2 Classification vs Regression

**Exercise 2.1** For each description: classification, regression, or neither? Identify a reasonable input variable(s) and output variable(s).

1. Predict tomorrow's maximum temperature from today's pressure, humidity and wind.
2. Identify which of five iris species a flower belongs to from four petal/sepal measurements.
3. Predict how many bikes will be rented in the next hour from weather and time of day.
4. Predict the star rating (from 1 to 5) a user will give a film.
5. Estimate the average salary of a company's employees from a random sample of 200 of them.
6. Flag credit card transactions as fraudulent ( 0.05% of transactions are fraud).
7. Predict whether a patient's tumor is benign or malignant from a biopsy image.
8. Predict a house's sale price from its size, location, and number of rooms.
9. Predict how many minutes a bus will be delayed given current traffic and weather data.
10. Group customers into segments based on purchasing behavior, without any predefined labels.